Lecture 13
Nonlinear Problems, Picard's method, Banach Fixed-Point Theorem, Newton-Raphson algorithm

Nonlinear Problems

Simple nonlinear (quasi-linear) PDE's

$$\frac{\partial u}{\partial t} = f(u, t)$$

Some famous *diffusion-reaction* equations:

- Fisher's equation (spreading of biological populations):

$$\frac{\partial u}{\partial t} = \Delta u + u(1 - u)$$

- Newell-Whitehead-Segel equation (Rayleigh-Bénard convection):

$$\frac{\partial u}{\partial t} = \Delta u + u(1 - u^2)$$

- Zeldovich equation (combustion theory):

$$\frac{\partial u}{\partial t} = \Delta u + u(1 - u)(u - \alpha)$$

Semi-discrete form of PDE's (systems of ODE's)

After FDM/FVM spatial discretization, nonlinear equations have the form:

$$\boldsymbol{u}' = \boldsymbol{f}(\boldsymbol{u}, t)$$

where $\boldsymbol{f}(\boldsymbol{u}, t) \in \mathbb{R}^n$ is a nonlinear vector function of $\boldsymbol{u} \in \mathbb{R}^n$.

- Fisher's equation:

$$\boldsymbol{u}' = -A\boldsymbol{u} + \boldsymbol{u} \circ (1 - \boldsymbol{u})$$

- Newell-Whitehead-Segel equation:

$$\boldsymbol{u}' = -A\boldsymbol{u} + \boldsymbol{u} \circ (1 - \boldsymbol{u} \circ \boldsymbol{u})$$

- Zeldovich equation:

$$\boldsymbol{u}' = -A\boldsymbol{u} + \boldsymbol{u} \circ (1 - \boldsymbol{u}) \circ (\boldsymbol{u} - \alpha)$$

Notation

Here, A denotes the (positive) Laplacian matrix and $\boldsymbol{u} \circ \boldsymbol{v}$ denotes the Hadamard (pointwise) product of two vectors:

$$\boldsymbol{u} \circ \boldsymbol{v} = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix} \circ \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix} = \begin{bmatrix} u_1 v_1 \\ u_2 v_2 \\ \vdots \\ u_n v_n \end{bmatrix}$$

So, that, for example, $\boldsymbol{u}^3 = \boldsymbol{u} \circ \boldsymbol{u} \circ \boldsymbol{u}$. Scalar addition:

$$\boldsymbol{u} + \alpha = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix} + \begin{bmatrix} \alpha \\ \alpha \\ \vdots \\ \alpha \end{bmatrix} = \begin{bmatrix} u_1 + \alpha \\ u_2 + \alpha \\ \vdots \\ u_n + \alpha \end{bmatrix}$$

Two-component reaction-diffusion systems

FitzHugh-Nagumo equation (activator-inhibitor system), two coupled nonlinear PDE's:

$$\begin{aligned}\frac{\partial u}{\partial t} &= D_u \Delta u + \lambda u - u^3 - \kappa - \sigma v \\ \tau \frac{\partial v}{\partial t} &= D_v \Delta v + u - v\end{aligned}$$

After discretization in space, two coupled systems of nonlinear ODE's:

$$\begin{aligned}\boldsymbol{u}' &= -D_u A \boldsymbol{u} + \lambda \boldsymbol{u} - \boldsymbol{u}^3 - \kappa - \sigma \boldsymbol{v} \\ \tau \boldsymbol{v}' &= -D_v A \boldsymbol{v} + \boldsymbol{u} - \boldsymbol{v}\end{aligned}$$

$$\frac{d}{dt}\begin{bmatrix}\boldsymbol{u}\\ \boldsymbol{v}\end{bmatrix} = \begin{bmatrix}-D_u A \boldsymbol{u} + \lambda \boldsymbol{u} - \boldsymbol{u}^3 - \kappa - \sigma \boldsymbol{v} \\ -\tau^{-1} D_v A \boldsymbol{v} + \tau^{-1} \boldsymbol{u} - \tau^{-1} \boldsymbol{v}\end{bmatrix} = \begin{bmatrix}\boldsymbol{f}_1(\boldsymbol{u}, \boldsymbol{v}) \\ \boldsymbol{f}_2(\boldsymbol{u}, \boldsymbol{v})\end{bmatrix}$$

Time-integration: Forward-Euler

The Forward-Euler time-integration method can be applied to any nonlinear system of ODE's of the form $\boldsymbol{u}' = \boldsymbol{f}(\boldsymbol{u}, t)$. FE iterations are:

$$\begin{aligned}\boldsymbol{u}(t_{k+1}) &= \boldsymbol{u}(t_k) + h_t\, \boldsymbol{f}(t_k, \boldsymbol{u}(t_k)), \quad k = 0, 1, \ldots \\ \boldsymbol{u}^{k+1} &= \boldsymbol{u}^k + h_t\, \boldsymbol{f}^k(\boldsymbol{u}^k), \quad k = 0, 1, \ldots\end{aligned}$$

For FE it does not matter if $\boldsymbol{f}^k(\boldsymbol{u}^k)$ is a *nonlinear* function $\boldsymbol{f} : \mathbb{R}^n \to \mathbb{R}^n$, but we know that FE may become unstable and requires sufficiently small time step h_t.

Forward-Euler algorithm for Fisher's equation

$$\boldsymbol{u}' = -A\boldsymbol{u} + \boldsymbol{u} - \boldsymbol{u}^2, \quad \boldsymbol{u}(t_0) = \boldsymbol{u}^0$$

Recognize that $\boldsymbol{f}(\boldsymbol{u}) = -A\boldsymbol{u} + \boldsymbol{u} - \boldsymbol{u}^2$.
Solving for $\boldsymbol{u}^k$ at t_k, $k = 0, 1, \ldots, K$, with $h_t = t_{k+1} - t_k$.

$$\begin{aligned}
&k = 0\\
&\text{while } k < K \text{ do :}\\
&\qquad \boldsymbol{u}^{k+1} = h_t\left[-A\boldsymbol{u}^k + \boldsymbol{u}^k - (\boldsymbol{u}^k)^2\right] + \boldsymbol{u}^k\\
&\qquad k = k + 1
\end{aligned}$$

Time-integration: Backward-Euler

Consider the Backward-Euler method applied to $\boldsymbol{u}' = \boldsymbol{f}(\boldsymbol{u}, t)$:

$$\boldsymbol{u}(t_{k+1}) = \boldsymbol{u}(t_k) + h_t\, \boldsymbol{f}(t_{k+1}, \boldsymbol{u}(t_{k+1})), \quad k = 0, 1, \ldots$$
$$\boldsymbol{u}^{k+1} = \boldsymbol{u}^k + h_t\, \boldsymbol{f}^{k+1}(\boldsymbol{u}^{k+1}), \quad k = 0, 1, \ldots$$

If $\boldsymbol{f}^k(\boldsymbol{u}^k)$ is a *nonlinear* function $\boldsymbol{f} : \mathbb{R}^n \to \mathbb{R}^n$, then to find $\boldsymbol{u}^{k+1}$ we need to solve the nonlinear problem

$$\boldsymbol{u}^{k+1} = h_t\, \boldsymbol{f}^{k+1}(\boldsymbol{u}^{k+1}) + \boldsymbol{u}^k$$

at each time step k. Here $\boldsymbol{u}^k$ is a known vector and $\boldsymbol{f}^{k+1}(\ldots)$ is a known function.

Backward-Euler method for Fisher's equation

After spatial discretization we have:

$$\boldsymbol{u}' = -A\boldsymbol{u} + \boldsymbol{u} - \boldsymbol{u}^2$$

Hence, $\boldsymbol{f}(\boldsymbol{u}) = -A\boldsymbol{u} + \boldsymbol{u} - \boldsymbol{u}^2$ and the FE iterations are

$$\boldsymbol{u}^{k+1} = h_t \left[-A\boldsymbol{u}^{k+1} + \boldsymbol{u}^{k+1} - (\boldsymbol{u}^{k+1})^2 \right] + \boldsymbol{u}^k, \quad k = 0, 1, \ldots$$

Must be solved for $\boldsymbol{u}^{k+1}$ at each iteration.

Picard's method

Iterative solution of nonlinear problems

Any nonlinear problem can be re-written as

$$\boldsymbol{u} = h\,\boldsymbol{f}(\boldsymbol{u}) + \boldsymbol{v}$$

For example:

$$\begin{aligned} \boldsymbol{u}^{k+1} &= h_t\left[-A\boldsymbol{u}^{k+1} + \boldsymbol{u}^{k+1} - (\boldsymbol{u}^{k+1})^2\right] + \boldsymbol{u}^k, \quad k = 0, 1, \ldots \\ \boldsymbol{u} &= h\,\boldsymbol{f}(\boldsymbol{u}) + \boldsymbol{v} \end{aligned}$$

We want to construct an *iterative* method:

$$\begin{aligned} &\boldsymbol{u}_0 \quad \text{some given initial guess} \\ &\boldsymbol{u}_{i+1} = \boldsymbol{p}(\boldsymbol{u}_i), \quad i = 0, 1, \ldots \end{aligned}$$

with the following properties:

- *Convergence:* $\lim_{i\to\infty} \boldsymbol{u}_i = \tilde{\boldsymbol{u}}$, where $\tilde{\boldsymbol{u}} = h\,\boldsymbol{f}(\tilde{\boldsymbol{u}}) + \boldsymbol{v}$
- *Small cost:* computing $\boldsymbol{p}(\boldsymbol{u}_i)$ is computationally not expensive
- *Fast rate of convergence:* progression to the limit is rapid

Picard's Method

The most simple choice of $\boldsymbol{p}(\boldsymbol{u})$ is

$$\boldsymbol{p}(\boldsymbol{u}) = h\,\boldsymbol{f}(\boldsymbol{u}) + \boldsymbol{v}$$

Picard's Method (functional iteration, fixed-point iteration):

1. choose initial guess $\boldsymbol{u}_0$
2. for $i = 0, 1, 2, \ldots$ do until convergence:

$$\boldsymbol{u}_{i+1} = h\,\boldsymbol{f}(\boldsymbol{u}_i) + \boldsymbol{v}$$

3. $\boldsymbol{u} = \boldsymbol{u}_{i+1}$

Time-stepping iterations (over k) are called *outer iterations*
Iterations of the nonlinear solver (over i) are called *inner iterations*
Measure of convergence: $\|h\,\boldsymbol{f}(\boldsymbol{u}_{i+1}) + \boldsymbol{v} - \boldsymbol{u}_{i+1}\|$ – error in the solution of the nonlinear equation.

Backward-Euler-Picard's algorithm for Fisher's equation

$$\boldsymbol{u}' = -A\boldsymbol{u} + \boldsymbol{u} - \boldsymbol{u}^2, \quad \boldsymbol{u}(t_0) = \boldsymbol{u}^0$$

Solving for $\boldsymbol{u}^k$ at t_k, $k = 0, 1, \ldots, K$, with $h_t = t_{k+1} - t_k$.

$$
\begin{aligned}
& k = 0 \\
& \text{while } k < K \text{ do :} \\
& \quad \boldsymbol{u}_0^{k+1} = \boldsymbol{u}^k \\
& \quad i = 0 \\
& \quad \text{while } \left\| h_t \left[-A\boldsymbol{u}_i^{k+1} + \boldsymbol{u}_i^{k+1} - (\boldsymbol{u}_i^{k+1})^2 \right] + \boldsymbol{u}^k - \boldsymbol{u}_i^{k+1} \right\| > \epsilon \text{ do :} \\
& \quad\quad \boldsymbol{u}_{i+1}^{k+1} = h_t \left[-A\boldsymbol{u}_i^{k+1} + \boldsymbol{u}_i^{k+1} - (\boldsymbol{u}_i^{k+1})^2 \right] + \boldsymbol{u}^k \\
& \quad\quad i = i + 1 \\
& \quad \boldsymbol{u}^{k+1} = \boldsymbol{u}_{i+1}^{k+1} \\
& \quad k = k + 1
\end{aligned}
$$

Banach Fixed-Point Theorem

Theorem: Let $h > 0$, $\boldsymbol{u}_0 \in \mathbb{R}^n$, and suppose that there exist numbers $\lambda \in (0, 1)$ and $\rho > 0$ such that

1. $\|\boldsymbol{f}(\boldsymbol{w}) - \boldsymbol{f}(\boldsymbol{u})\| \leq \frac{\lambda}{h}\|\boldsymbol{w} - \boldsymbol{u}\|$ for every $\boldsymbol{w}, \boldsymbol{u} \in \mathcal{B}_\rho(\boldsymbol{u}_0)$, where $\|\cdot\|$ is a vector norm, and $\mathcal{B}_\rho(\boldsymbol{u}_0) := \{\boldsymbol{u} \in \mathbb{R}^n \,|\, \|\boldsymbol{u} - \boldsymbol{u}_0\| \leq \rho\}$ is the closed ball of radius ρ centered at $\boldsymbol{u}_0$
2. $\boldsymbol{u}_1 \in \mathcal{B}_{(1-\lambda)\rho}(\boldsymbol{u}_0)$

Then, for the iterations $\boldsymbol{u}_{i+1} = h\,\boldsymbol{f}(\boldsymbol{u}_i) + \boldsymbol{v}$, $i = 0, 1, \ldots$ the following holds:

a. $\boldsymbol{u}_i \in \mathcal{B}_\rho(\boldsymbol{u}_0)$ for every $i = 0, 1, \ldots$
b. $\tilde{\boldsymbol{u}} : \lim_{i\to\infty} \boldsymbol{u}_i$ exists, satisfies equation $\tilde{\boldsymbol{u}} = h\,\boldsymbol{f}(\tilde{\boldsymbol{u}}) + \boldsymbol{v}$, and $\tilde{\boldsymbol{u}} \in \mathcal{B}_\rho(\boldsymbol{u}_0)$
c. no other point in $\mathcal{B}_\rho(\boldsymbol{u}_0)$ is a solution of $\boldsymbol{u} = h\,\boldsymbol{f}(\boldsymbol{u}) + \boldsymbol{v}$

Proof (1)

Proof: We use induction to show that

$$\|\boldsymbol{u}_{i+1} - \boldsymbol{u}_i\| \le \lambda^i (1-\lambda)\rho$$

and that $\boldsymbol{u}_{i+1} \in \mathcal{B}_\rho(\boldsymbol{u}_0)$ for all $i = 0, 1, \dots$ It is true for $i = 0$ due to (ii), i.e., $\|\boldsymbol{u}_1 - \boldsymbol{u}_0\| \le (1-\lambda)\rho$. Assume it is true for all $m = 0, 1, \dots, i-1$, i.e., $\|\boldsymbol{u}_i - \boldsymbol{u}_{i-1}\| \le \lambda^{i-1}(1-\lambda)\rho$. Then, due to (i):

$$\begin{aligned}\|\boldsymbol{u}_{i+1} - \boldsymbol{u}_i\| &= \|[h\boldsymbol{f}(\boldsymbol{u}_i) + \boldsymbol{v}] - [h\boldsymbol{f}(\boldsymbol{u}_{i-1}) + \boldsymbol{v}]\| = h\|\boldsymbol{f}(\boldsymbol{u}_i) - \boldsymbol{f}(\boldsymbol{u}_{i-1})\| \\ &\le \lambda\|\boldsymbol{u}_i - \boldsymbol{u}_{i-1}\| \le \lambda\lambda^{i-1}(1-\lambda)\rho = \lambda^i(1-\lambda)\rho\end{aligned}$$

This carries the induction from $i-1$ to i. Consider the sum:

$$\boldsymbol{u}_{i+1} - \boldsymbol{u}_0 = \boldsymbol{u}_{i+1} - \boldsymbol{u}_i + \boldsymbol{u}_i - \boldsymbol{u}_{i-1} + \dots + \boldsymbol{u}_1 - \boldsymbol{u}_0 = \sum_{j=0}^{i} (\boldsymbol{u}_{j+1} - \boldsymbol{u}_j)$$

By triangle inequality for a vector norm

$$\|\boldsymbol{u}_{i+1} - \boldsymbol{u}_0\| = \left\| \sum_{j=0}^{i} (\boldsymbol{u}_{j+1} - \boldsymbol{u}_j) \right\| \le \sum_{j=0}^{i} \|\boldsymbol{u}_{j+1} - \boldsymbol{u}_j\|, \quad i = 0, 1, \dots$$

Proof (2)

Exploiting $\|\boldsymbol{u}_{i+1} - \boldsymbol{u}_i\| \leq \lambda^i(1-\lambda)\rho$ and the sum of a geometric series $\sum_{j=0}^{i} \lambda^j = (1-\lambda^{i+1})/(1-\lambda)$,

$$\|\boldsymbol{u}_{i+1} - \boldsymbol{u}_0\| \leq \sum_{j=0}^{i} \lambda^j (1-\lambda)\rho = (1-\lambda^{i+1})\rho \leq \rho, \quad i = 0, 1, \ldots$$

Hence, $\boldsymbol{u}_{i+1} \in \mathcal{B}_\rho(\boldsymbol{u}_0)$ and (a) is true.

Now consider:

$$\begin{aligned}\|\boldsymbol{u}_{i+k} - \boldsymbol{u}_i\| &= \left\|\sum_{j=0}^{k-1}(\boldsymbol{u}_{i+j+1} - \boldsymbol{u}_{i+j})\right\| \leq \sum_{j=0}^{k-1} \|\boldsymbol{u}_{i+j+1} - \boldsymbol{u}_{i+j})\| \\ &\leq \sum_{j=0}^{k-1} \lambda^{i+j}(1-\lambda)\rho = \lambda^i(1-\lambda^k)\rho, \quad i = 0, 1, \ldots; \quad k = 1, 2, \ldots\end{aligned}$$

Proof (3)

Since $\lambda \in (0,1)$ and $\|\boldsymbol{u}_{i+k} - \boldsymbol{u}_i\| \leq \lambda^i(1-\lambda^k)\rho$, for any $\epsilon > 0$ we may chose N large enough so that that

$$\|\boldsymbol{u}_{i+k} - \boldsymbol{u}_i\| < \epsilon$$

for all $i, (i+k) > N$. In other words, $\{\boldsymbol{u}_i\}$ is a Cauchy sequence. Since the set $\mathcal{B}_\rho(\boldsymbol{u}_0)$ is compact (closed and bounded), the Cauchy sequence $\{\boldsymbol{u}_i\}$ converges to a limit within this set. This proves the existence of $\lim_{i\to\infty} \boldsymbol{u}_i = \tilde{\boldsymbol{u}} \in \mathcal{B}_\rho(\boldsymbol{u}_0)$.

Assume that there exists another $\boldsymbol{u}^* \in \mathcal{B}_\rho(\boldsymbol{u}_0)$, $\boldsymbol{u}^* \neq \tilde{\boldsymbol{u}}$, such that $\boldsymbol{u}^* = h\boldsymbol{f}(\boldsymbol{u}^*) + \boldsymbol{v}$. In which case $\|\boldsymbol{u}^* - \tilde{\boldsymbol{u}}\| > 0$. On the other hand,

$$\begin{aligned}\|\boldsymbol{u}^* - \tilde{\boldsymbol{u}}\| &= \|[h\boldsymbol{f}(\boldsymbol{u}^*) + \boldsymbol{v}] - [h\boldsymbol{f}(\tilde{\boldsymbol{u}}) + \boldsymbol{v}]\| = h\|\boldsymbol{f}(\boldsymbol{u}^*) - \boldsymbol{f}(\tilde{\boldsymbol{u}})\| \\ &\leq \lambda\|\boldsymbol{u}^* - \tilde{\boldsymbol{u}}\| < \|\boldsymbol{u}^* - \tilde{\boldsymbol{u}}\|, \quad \text{i.e.} \quad \|\boldsymbol{u}^* - \tilde{\boldsymbol{u}}\| < \|\boldsymbol{u}^* - \tilde{\boldsymbol{u}}\|\end{aligned}$$

which is not possible. Hence, $\|\boldsymbol{u}^* - \tilde{\boldsymbol{u}}\| = 0$ and $\tilde{\boldsymbol{u}}$ is unique in $\mathcal{B}_\rho(\boldsymbol{u}_0)$. $\square$

Inner vs outer iterations

- In the outer iterations of the Backward Euler Method:

$$\boldsymbol{u}^{k+1} = h_t \boldsymbol{f}^{k+1}(\boldsymbol{u}^{k+1}) + \boldsymbol{u}^k$$

  it is tempting to take a large h_t
- In the inner iterations of the Picard Method:

$$\boldsymbol{u}_{i+1}^{k+1} = h_t \boldsymbol{f}^{k+1}(\boldsymbol{u}_i^{k+1}) + \boldsymbol{u}^k$$

  it is often required to have a small $h = h_t$. Why? Suppose, $\|\boldsymbol{f}(\boldsymbol{u}) - \boldsymbol{f}(\boldsymbol{w})\| \leq \xi \|\boldsymbol{u} - \boldsymbol{w}\|$ for all $\boldsymbol{u}, \boldsymbol{w} \in \mathcal{B}_\rho(\boldsymbol{u}_0)$, where $\xi > 1$. We can always choose $h_t > 0$ small enough that $\xi = \lambda / h_t$ with $\lambda \in (0, 1)$
- Approach: start with some reasonable h_t. If inner iterations converge very slowly (more than 10-20 iterations), disregard this h_t and restart inner iterations with a smaller h_t
- It is important to have a good initial guess $\boldsymbol{u}_0^{k+1}$, e.g. $\boldsymbol{u}_0^{k+1} = \boldsymbol{u}^k$

Newton-Raphson algorithm

Idea behind the Newton-Raphson method

We want to solve

$$\boldsymbol{u} = h\boldsymbol{f}(\boldsymbol{u}) + \boldsymbol{v}$$

We have some initial guess $\boldsymbol{u}_i$. Ideally, to get to $\boldsymbol{u}$ from $\boldsymbol{u}_i$, we need to know $\boldsymbol{u} - \boldsymbol{u}_i$, since

$$\boldsymbol{u} = \boldsymbol{u}_i + \boldsymbol{u} - \boldsymbol{u}_i$$

Newton-Raphson's algorithm finds an approximation for $\boldsymbol{u} - \boldsymbol{u}_i$

$$\boldsymbol{u} - \boldsymbol{u}_i \approx \boldsymbol{p}_i(\boldsymbol{u}_i)$$

and iterates as follows:

$$\boldsymbol{u}_{i+1} = \boldsymbol{u}_i + \boldsymbol{p}_i(\boldsymbol{u}_i)$$

Compare this to a fixed-point iteration (Picard's method): $\boldsymbol{u}_{i+1} = \boldsymbol{p}(\boldsymbol{u}_i)$.

Derivation via Taylor's expansion

$$\boldsymbol{u} = h\boldsymbol{f}(\boldsymbol{u}) + \boldsymbol{v},$$

$$\boldsymbol{f}(\boldsymbol{u}) = \boldsymbol{f}(\boldsymbol{u}_i) + \left.\frac{\partial \boldsymbol{f}(\boldsymbol{u})}{\partial \boldsymbol{u}}\right|_{\boldsymbol{u}=\boldsymbol{u}_i} (\boldsymbol{u} - \boldsymbol{u}_i) + \mathcal{O}(\|\boldsymbol{u} - \boldsymbol{u}_i\|^2),$$

$$\boldsymbol{u} = h\boldsymbol{f}(\boldsymbol{u}_i) + h \left.\frac{\partial \boldsymbol{f}(\boldsymbol{u})}{\partial \boldsymbol{u}}\right|_{\boldsymbol{u}=\boldsymbol{u}_i} (\boldsymbol{u} - \boldsymbol{u}_i) + \mathcal{O}(h\|\boldsymbol{u} - \boldsymbol{u}_i\|^2) + \boldsymbol{v}$$

$$\boldsymbol{u} - \boldsymbol{u}_i = h\boldsymbol{f}(\boldsymbol{u}_i) + h \left.\frac{\partial \boldsymbol{f}(\boldsymbol{u})}{\partial \boldsymbol{u}}\right|_{\boldsymbol{u}=\boldsymbol{u}_i} (\boldsymbol{u} - \boldsymbol{u}_i) + \mathcal{O}(h\|\boldsymbol{u} - \boldsymbol{u}_i\|^2) + \boldsymbol{v} - \boldsymbol{u}_i$$

$$\boldsymbol{u} - \boldsymbol{u}_i - h \left.\frac{\partial \boldsymbol{f}(\boldsymbol{u})}{\partial \boldsymbol{u}}\right|_{\boldsymbol{u}=\boldsymbol{u}_i} (\boldsymbol{u} - \boldsymbol{u}_i) = h\boldsymbol{f}(\boldsymbol{u}_i) + \boldsymbol{v} - \boldsymbol{u}_i + \mathcal{O}(h\|\boldsymbol{u} - \boldsymbol{u}_i\|^2)$$

$$\left[I - h \left.\frac{\partial \boldsymbol{f}(\boldsymbol{u})}{\partial \boldsymbol{u}}\right|_{\boldsymbol{u}=\boldsymbol{u}_i}\right] (\boldsymbol{u} - \boldsymbol{u}_i) \approx \boldsymbol{v} + h\,\boldsymbol{f}(\boldsymbol{u}_i) - \boldsymbol{u}_i$$

$$\boldsymbol{u} - \boldsymbol{u}_i \approx \left[I - h \left.\frac{\partial \boldsymbol{f}(\boldsymbol{u})}{\partial \boldsymbol{u}}\right|_{\boldsymbol{u}=\boldsymbol{u}_i}\right]^{-1} [\boldsymbol{v} + h\,\boldsymbol{f}(\boldsymbol{u}_i) - \boldsymbol{u}_i]$$

Jacobian

Newton-Raphson's iterations are

$$\boldsymbol{u}_{i+1} = \boldsymbol{u}_i + \boldsymbol{p}_i(\boldsymbol{u}_i)$$

with

$$\boldsymbol{p}_i(\boldsymbol{u}_i) = [I - hJ(\boldsymbol{u}_i)]^{-1} [\boldsymbol{v} + h\,\boldsymbol{f}(\boldsymbol{u}_i) - \boldsymbol{u}_i]$$

If $\boldsymbol{u} \in \mathbb{R}^n$ and $\boldsymbol{f}(\boldsymbol{u}) : \mathbb{R}^n \to \mathbb{R}^n$, then the Jacobian matrix $J \in \mathbb{R}^{n \times n}$ is defined as

$$J(\boldsymbol{u}_i) = \left.\frac{\partial \boldsymbol{f}(\boldsymbol{u})}{\partial \boldsymbol{u}}\right|_{\boldsymbol{u}=\boldsymbol{u}_i}$$

$$[J(\boldsymbol{u}_i)]_{qr} = \left.\frac{\partial f_q(\boldsymbol{u})}{\partial u_r}\right|_{\boldsymbol{u}=\boldsymbol{u}_i}$$

Convergence and modifications

- As long as $h > 0$ is sufficiently small, Newton-Raphson's algorithm has quadratic convergence, i.e., there exists a constant $c > 0$, such that, for sufficiently large i,

$$\|\tilde{\boldsymbol{u}} - \boldsymbol{u}_{i+1}\| \leq c\|\tilde{\boldsymbol{u}} - \boldsymbol{u}_i\|^2$$

- With $J = 0$, we get Picard's algorithm:

$$\boldsymbol{u}_{i+1} = \boldsymbol{u}_i + [I - h0]^{-1} [\boldsymbol{v} + h\,\boldsymbol{f}(\boldsymbol{u}_i) - \boldsymbol{u}_i] = h\,\boldsymbol{f}(\boldsymbol{u}_i) + \boldsymbol{v}$$

- With $J(\boldsymbol{u}_i) \approx J(\boldsymbol{u}_0)$ we get Modified Newton-Raphson's algorithm:

$$\boldsymbol{u}_{i+1} = \boldsymbol{u}_i + [I - hJ(\boldsymbol{u}_0)]^{-1} [\boldsymbol{v} + h\,\boldsymbol{f}(\boldsymbol{u}_i) - \boldsymbol{u}_i]$$

Backward-Euler-Newton-Raphson's algorithm for Fisher's equation

$$\boldsymbol{u}' = -A\boldsymbol{u} + \boldsymbol{u} - \boldsymbol{u}^2, \quad \boldsymbol{u}(t_0) = \boldsymbol{u}^0$$

Solving for $\boldsymbol{u}^k$ at t_k, $k = 0, 1, \ldots, K$, with $h_t = t_{k+1} - t_k$; $\boldsymbol{f}(\boldsymbol{u}) = -A\boldsymbol{u} + \boldsymbol{u} - \boldsymbol{u}^2$, and $\boldsymbol{v} = \boldsymbol{u}^k$

$$
\begin{aligned}
&k = 0\\
&\text{while } k < K \text{ do :}\\
&\qquad \boldsymbol{u}_0^{k+1} = \boldsymbol{u}^k\\
&\qquad i = 0\\
&\qquad \text{while } \|\boldsymbol{u}^k + h_t\boldsymbol{f}(\boldsymbol{u}_i^{k+1}) - \boldsymbol{u}_i^{k+1}\| > \epsilon \text{ do :}\\
&\qquad\qquad \boldsymbol{u}_{i+1}^{k+1} = \boldsymbol{u}_i^{k+1} + \left[I - h_t J(\boldsymbol{u}_i^{k+1})\right]^{-1}\left[\boldsymbol{u}^k + h_t\,\boldsymbol{f}(\boldsymbol{u}_i^{k+1}) - \boldsymbol{u}_i^{k+1}\right]\\
&\qquad\qquad i = i + 1\\
&\qquad \boldsymbol{u}^{k+1} = \boldsymbol{u}_{i+1}^{k+1}\\
&\qquad k = k + 1
\end{aligned}
$$

Jacobian matrix of the Fisher's equation

$$
\begin{aligned}
\boldsymbol{f}(u) &= -A\boldsymbol{u} + \boldsymbol{u} - \boldsymbol{u}^2 \\
[J(\boldsymbol{u})]_{q,r} &= \frac{\partial f_q(\boldsymbol{u})}{\partial u_r} = \frac{\partial}{\partial u_r}\left[-\sum_{j=1}^{n} a_{q,j}u_j + u_q - u_q^2\right] \\
&= -\sum_{j=1}^{n} a_{q,j}\frac{\partial u_j}{\partial u_r} + \frac{\partial u_q}{\partial u_r} - 2u_q\frac{\partial u_q}{\partial u_r} \\
&= -\sum_{j=1}^{n} a_{q,j}\delta_{j,r} + \delta_{qr} - 2u_q\delta_{qr} \\
&= -a_{qr} + \delta_{qr} - 2u_q\delta_{qr} \\
J(\boldsymbol{u}_i^{k+1}) &= -A + I - 2\,\mathrm{diag}(\boldsymbol{u}_i^{k+1})
\end{aligned}
$$

where $\delta_{q,r}$ is the Kronecker delta function, $I \in \mathbb{R}^{n\times n}$ is the identity matrix, and $\mathrm{diag}(\boldsymbol{u}_i^{k+1})$ is the matrix with the vector $\boldsymbol{u}_i^{k+1}$ on its diagonal.

Inner iterations require a linear solver

To compute the update vector

$$\boldsymbol{p}_i(\boldsymbol{u}_i^{k+1}) = \left[I - h_t J(\boldsymbol{u}_i^{k+1})\right]^{-1} \left[\boldsymbol{u}^k + h_t\, \boldsymbol{f}(\boldsymbol{u}_i^{k+1}) - \boldsymbol{u}_i^{k+1}\right]$$

at each inner iteration we solve the linear system

$$\left[I - h_t J(\boldsymbol{u}_i^{k+1})\right] \boldsymbol{p}_i(\boldsymbol{u}_i^{k+1}) = \boldsymbol{u}^k + h_t\, \boldsymbol{f}(\boldsymbol{u}_i^{k+1}) - \boldsymbol{u}_i^{k+1}$$

for $\boldsymbol{p}_i(\boldsymbol{u}_i^{k+1})$ and then compute the Newton-Raphson update as:

$$\boldsymbol{u}_{i+1}^{k+1} = \boldsymbol{u}_i^{k+1} + \boldsymbol{p}_i(\boldsymbol{u}_i^{k+1})$$